

思考6.

要满足 D 可解析. $z^2 a^2 \neq 0$. $z+a \neq 0$.

$$\frac{1}{z^2 a^2} = \frac{1}{2a} \left(\frac{1}{z-a} - \frac{1}{z+a} \right)$$

设 $z = x+iy$, $a = b+ic$

$$\frac{1}{z-a} = \frac{(x-b)-i(y-c)}{(x-b)^2 + (y-c)^2} \quad \text{显然可解析}$$

同理 $\frac{1}{z+a}$ 可解析. 关键在于 $\pm a$ 是否在 C 内部

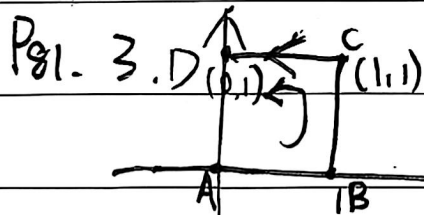
任取 $-HS = \left[\oint \frac{1}{z-a} dz - \oint \frac{1}{z+a} dz \right] \frac{1}{2a}$
 $\pm a$ 在 C 外 or $\pm a$ 在 C 内.

$$\oint_C \frac{dz}{z-a} \quad n=1$$

$$= 2\pi i$$

a 在外, $-a$ 在内 or a 在内, $-a$ 在外

$$-\frac{\pi i}{a} \quad a \text{ 在外, } -a \text{ 在内}$$



$AB: x=t \in (0,1), y=0$

$\therefore LHS = \int_0^1 \cos\left(3 + \frac{1}{t-3}\right) dt, y=1$

$CD: x=t \in (0,1), y=0, LHS = \int_1^0 \cos\left(3 + \frac{1}{t-3}\right) dt$

$$\left(\int_{AB} + \int_{CD} \right) \cos\left(3 + \frac{1}{z-3}\right) dz = \oint_C \cos\left(3 + \frac{1}{z-3}\right) dz$$

$\cos\left(3 + \frac{1}{z-3}\right)$ 在 $C \setminus \{3\}$ 上可解析

$$\therefore \oint_C \cos\left(3 + \frac{1}{z-3}\right) dz = 0$$

Pr. 4. $z = z_0 + re^{i\theta}$

P81. 4.

$$\text{令 } z = z_0 + re^{i\theta} \quad \theta \in [0, 2\pi]$$

$$dz = ire^{i\theta} d\theta \Rightarrow d\theta = -i \frac{1}{r} e^{-i\theta} dz$$

$$\int_0^{2\pi} f(z) \cdot e^{ik\theta} (-i) \frac{1}{r} e^{-i\theta} dz$$

$$\text{LHS} = \int f(z) e^{i(k-1)\theta} \frac{-i}{r} d(z_0 + re^{i\theta}) = \int e^{i(k-1)\theta + \frac{3\pi}{2}} f(z) dz$$

$$\text{LHS} = \oint_{\substack{\text{以 } z_0 \text{ 为中心} \\ r \text{ 为半径圆}}} f(z) e^{i(k-1)\theta} d e^{i\theta} = i \oint \frac{f(z)}{\left(\frac{z-z_0}{r}\right)^{1-k}} d\left(\frac{z-z_0}{r}\right) = -\frac{i}{r^k} \oint \frac{f(z)}{\left(\frac{z-z_0}{r}\right)^{1-k}}$$

$$\text{当 } k=1, \text{ LHS} = -\frac{i}{r} \oint f(z) dz = 0$$

$$k \geq 1, \quad -\frac{i}{r^k} \oint f(z) (z-z_0)^{k-1} dz = 0$$

$$\therefore k \geq 1 \text{ 时, LHS} = 0$$

$$7. (1) \oint_0^{2\pi} \frac{-\sin t + i z \cos t}{\cos t + i z \sin t} dt$$

(2) $\frac{1}{z}$ 在 $\mathbb{C} \setminus \{0\}$ 上解析.

$$\oint_C \frac{1}{z} dz = \oint_{C_1} \frac{1}{z} dz \quad \text{其中 } C_1 \text{ 为 } 0 \text{ 为圆心, } R=1 \text{ 的圆}$$

$$(2) \text{ 同理 } z^2 \neq 0 \quad \oint_{C_1} \frac{1}{z} dz = \int_0^{2\pi} \frac{-\sin \theta}{\cos \theta} d\theta = 0 \quad 2\pi i$$

$$\oint_C \frac{1}{z^2} dz = \oint_{C_1} \frac{1}{z^2} dz \quad z = e^{i\theta} \quad \theta \in (-\pi, \pi)$$

$$\therefore \oint_{C_1} \frac{1}{z^2} dz = \int_{-\pi}^{\pi} \frac{-\sin \theta}{\cos^2 \theta} d\theta = \int_{-\pi}^{\pi} \frac{\sin \theta}{\sin^2 \theta - 1} d\theta = 0$$

14) $x = 1 + \cos \theta, y = \sin \theta, \theta \in (0, 2\pi)$

$\therefore \text{LHS} = \oint_C \frac{1}{z-1} - \frac{1}{z+1} dz = \oint_C \frac{1}{z-1} dz$

$\therefore \frac{1}{z-1}$ 在 $C/1$ 上解析 $= \frac{1}{2} \int_{-\pi}^{\pi} \frac{-\sin \theta}{\cos \theta} d\theta = 0$

$\therefore \oint_C \frac{1}{z-1} dz = 0$

(6) $\frac{1}{1-z^3}$ 在 $C/1$ 上解析, $1 \notin C$

$\therefore \oint \frac{1}{1-z^3} dz = 0$

21. $\frac{e^z}{z}$ 在 $C/0$ 上解析

$\therefore \oint_{|z|=1} \frac{e^z}{z} dz = 0$

取 $z = \cos \theta \in (0, 2\pi)$

(2) $\int_0^{2\pi} \frac{e^z}{z} dz = \int_0^{2\pi} e^{\cos \theta} \frac{-\sin \theta}{\cos \theta} d\theta =$

$e^{\cos \theta} \cos(\sin \theta) d\theta = \frac{e^{\cos \theta}}{\cos \theta} \cos \theta \cos(\sin \theta) d\theta = \frac{e^{\cos \theta}}{\cos \theta} d\sin(\sin \theta)$

$\oint \frac{e^z}{z} dz = 2\pi i e^z \Big|_{z=0} = 2\pi i$

令 $z = e^{i\theta} \quad \theta \in (0, 2\pi) (-\pi, \pi)$

$\text{LHS} = \int_{-\pi}^{\pi} \frac{e^{e^{i\theta}}}{e^{i\theta}} de^{i\theta} = \int_{-\pi}^{\pi} i e^{\cos \theta + i \sin \theta} d\theta = i \int_{-\pi}^{\pi} e^{\cos \theta} [\cos(\sin \theta) + i \sin(\sin \theta)] d\theta$

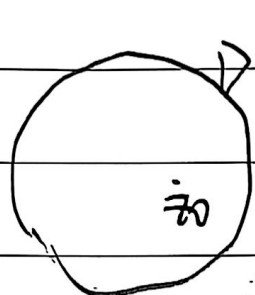
$= \int_{-\pi}^{\pi} e^{\cos \theta} [i \cos(\sin \theta) - \sin(\sin \theta)] d\theta = 2\pi i$

$\Rightarrow \int_{-\pi}^{\pi} e^{\cos \theta} \underbrace{\cos \theta \cos(\sin \theta)}_{\text{偶}} d\theta = 2\pi$

$\Rightarrow \int_0^{\pi} e^{\cos \theta} \cos \theta \cos(\sin \theta) d\theta = \pi$

1b. $f(z)$ 在 $D \setminus \{z_0\}$ 解析

$\lim_{z \rightarrow z_0} f(z) = A$ 由形变定理:



$$\oint_{C_1} f(z) dz = \oint_{C_2} f(z) dz$$

取 $C_2: |z - z_0| < C_8$ $C_8 \rightarrow 0 \Rightarrow \lim_{z \rightarrow z_0} |z - z_0| = 0$

$$\therefore z = z_0 + \alpha, \quad \lim_{z \rightarrow z_0} \alpha = 0$$

$$\text{令 } z = z_0 + C_8 e^{i\theta} \quad \theta \in (0, 2\pi)$$

$$\begin{aligned} \therefore \text{LHS} &= \int_0^{2\pi} f(z_0 + C_8 e^{i\theta}) C_8 i e^{i\theta} d\theta \\ &= C_8 i \int_0^{2\pi} f(z_0 + C_8 e^{i\theta}) e^{i\theta} d\theta. \end{aligned}$$

$\therefore f(z)$ 在 z_0 旁有界

$$\therefore \left| \frac{\text{LHS}}{C_8} \right| = \left| \int_0^{2\pi} f(z_0 + C_8 e^{i\theta}) i e^{i\theta} d\theta \right| < \varepsilon, \quad \varepsilon < \varepsilon_0$$

$$\therefore \text{LHS} = \oint_C f(z) dz = 0$$

$$\oint_{C_8} f(z) dz \leq M \cdot 2\pi C_8 \quad \text{形变公式}$$